

TREAT

CLEAN

PROTECT

DRY

Treat

Loosen and lift soil with chemistry and time before anything touches the vehicle.

EXIT STANDARD
Full, even chemical coverage on every panel, wheel and tire, with soil visibly breaking down before the car reaches friction.

KEY CONTROLS

CONTROL	TARGET	HOW WE VERIFY
Chemical sequence	High-pH presoak first, then low-pH step	Alkaline breaks down road film; acidic neutralizes it and lifts mineral soil.
Concentration	Product data-sheet dilution, confirmed by titration	Weekly titration of each product. Supply water pH and hardness shift results, so don't trust the injector setting alone.
Dwell	~4–6 ft of travel from presoak to the next step (≈9 sec at a 100 CPH setting)	Calculated per site from chain speed and arch spacing.
Chain speed	Fastest speed that still passes the test car, held steady all day	Hourly setpoint log. Speeding up at peak and slowing off-peak makes quality uneven.
Nozzle coverage	Every nozzle flowing, even pattern, no dry stripes	Hourly visual check; timed flow test monthly. Replace nozzles above ~110% of rated flow.
Wheels & tires	Wheel chemistry plus high-pressure blasting before friction	Test car shows no brake dust at exit.
Temperature	Warm supply water in cold weather	Seasonal set-up check. Warm the water, not the chemical drum.

FAILURE → WHAT THE CUSTOMER SEES

Low concentration or short dwell

Road-film haze on sides and rear

Plugged or misaligned nozzle

Same dirty stripe on every car

Over-concentration

Soapy film and streaks, worst on light colors

Empty drum or lost prime

Whole step missing; dull, dirty finish

Worn nozzles

Nothing visible; up to 30% wasted chemical

CHECKS

DAILY

- Drum levels; pumps primed
- All presoak arches spraying evenly

WEEKLY

- Titrate every product; adjust injectors

MONTHLY

- Timed flow test on every nozzle

MEASURE

- Chemical use per car, by product, vs. target
- Chain speed actual vs. setpoint
- Drum-level and no-flow alarms

Rule: When speed cuts dwell, apply more liquid and rely on friction. Don't raise concentration.

TREAT

CLEAN

PROTECT

DRY

Clean

Remove the loosened soil with friction and rinse, without marking or damaging the vehicle.

EXIT STANDARD
No film, dirt bands, rocker-panel dirt or brake dust. No swirls, grease marks or damage. Rinsed to neutral pH.

KEY CONTROLS

CONTROL	TARGET	HOW WE VERIFY
Brush speed	Within the equipment's RPM spec (foam runs faster than cloth)	Weekly check. Outside spec risks vehicle or equipment damage.
Contact pressure	Cloth presses ~3–4 in into the vehicle; never enough to fold mirrors	Monthly penetration measurement. Less than 3 in means worn media.
Media condition	No tears, matting or trapped grit	Daily visual; weekly hands-on. Replace on wear and cars washed, not age.
Lubrication	Foaming brush detergent on the car before and during friction	Hourly walk: brushes wet and foamed.
Contamination	Zero oil or grease on the media	Daily check. Grease on cloth transfers to paint.
Wrap retract	Retracts as the vehicle passes the front quarter panel	Watch wraps on a test car; mirrors are the top damage claim.
Car spacing	3.5–7 ft between cars	Gives wraps room to reset between cars.
Rinse	Full high-pressure rinse coverage; ~pH 7 at exit	Quarterly pH check on a test car.

FAILURE → WHAT THE CUSTOMER SEES

Grit in media / no lubrication

Swirl marks and fine scratches

Pressure too high / late retract

Folded or broken mirrors

Worn or mis-set rocker brush

Dirty lower panels

Torn media at one height

Repeating dirty band

Grease on media

Black marks on paint

Poor rinse

Chemical film and streaks

CHECKS

DAILY

- Test car before opening
- Media visual: tears, grease

WEEKLY

- Hands-on media check: fraying (cloth), cracks (foam)
- Brush speed and pressure settings

MONTHLY

- Deep clean; check for rust and loose parts; measure penetration

MEASURE

- Brush motor load vs. each brush's baseline
- Cars washed since last media change
- Damage claims per 1,000 cars

Rule: Most damage claims trace to pressure and retract timing. Keep entrance and exit cameras running for every claim.

Protect

Seal the clean surface, rinse it spot-free and finish the tires. This is the shine the customer pays for.

EXIT STANDARD

Uniform gloss, water beading evenly, no spots or streaks. Tires evenly dressed with no sling on wheels or paint.

KEY CONTROLS

CONTROL	TARGET	HOW WE VERIFY
Sequence	Protectant / ceramic → drying agent → spot-free final rinse	A protectant can't fix poor rinsing upstream.
Protectant dose	Full rated dose	Usage per car vs. target. An under-dosed ceramic still beads water but gives little shine or durability.
Final rinse water	Dissolved solids under 10 ppm (under 20 acceptable; 5 or less fully spot-free)	Weekly reading on the RO output, continuous if a sensor is installed.
Reclaim water	Never in the final rinse	Plumbing check. Reclaim carries soap, dirt and minerals.
RO feed	Softened ahead of the membranes	Scale on membranes drives output out of spec.
Tire shine dose	~0.8–0.9 oz per car, split front and rear tire	Usage per car; test car shows even dressing.
Tire shine applicator	Level, clean, shielded from dryer air, on rubber only	A tilted manifold dumps product to one side.

FAILURE → WHAT THE CUSTOMER SEES

High dissolved solids in rinse

Water spots, worst on dark cars in sun

Under-dosed protectant

Looks fine at exit; shine fades fast

Tire shine overdose or tilted manifold

Sling on wheels and lower panels

Protectant line empty

Dull finish; water sheets, car exits wet

Reclaim crossover

Spots and film after drying

CHECKS

DAILY

- Test car: gloss, beading, tire shine
- Protectant and tire shine levels

WEEKLY

- RO output reading logged
- Tire shine manifold level and clean

QUARTERLY

- Full water panel: pH, hardness, dissolved solids, chlorine

MEASURE

- Protectant and tire shine use per car vs. target
- RO output ppm and recovery ratio
- Spot and shine complaints per 1,000 cars

Rule: Beading alone doesn't prove protection. Confirm the dose per car.

TREAT

CLEAN

PROTECT

DRY

Dry

Strip the water off before the car leaves. Drying is mostly decided upstream: water that beads blows off, water that sheets doesn't.

EXIT STANDARD

Roof, hood, glass, mirrors and rear hatch free of standing water, with no re-wetting after the dryers.

KEY CONTROLS

CONTROL	TARGET	HOW WE VERIFY
Water condition	Water beading as the car reaches the dryers	Watch the roof at the dryer entrance. Sheetting water points to the drying agent, not the blowers.
Placement	First overhead blowers remove the bulk; the rest angled to sweep one way	Opposing airflows push water back onto dried areas.
Rear of SUVs and vans	Water pushed down and off the back, not toward it	Test with an SUV; the rear hatch is the most common wet spot.
Rinse spacing	No high-pressure rinse mist reaching the dryer zone	Visual on a running tunnel.
Nozzle distance	Close to the body (~6–8 in on contouring side nozzles)	Air speed drops fast with distance.
Airflow path	Clean intake and safety screens; no torn nozzles or air leaks	Weekly screen cleaning; leaks fixed immediately.
Motor health	Amps within each blower's baseline; no noise or vibration	Low amps point to a blocked intake. Noise or vibration means bearings or impeller balance.
Chain speed	Leaves enough time under the dryers	Test-car dryness checked at peak speed.

FAILURE → WHAT THE CUSTOMER SEES

Drying agent missing or weak

Wet roof and hood; spots later

Rear blower angle wrong

Wet rear hatch on SUVs

Opposing flows / rinse mist

Car re-wets after drying

Blocked intake / torn nozzle

Gradually wetter cars, energy wasted

Speed too high

Wet cars at peak hours only

CHECKS

HOURLY

- All blowers running; exit cars dry

WEEKLY

- Clean intake and safety screens
- Inspect nozzles and air bags for tears

MONTHLY

- Amp reading on every blower vs. baseline
- Listen for bearing noise and vibration

MEASURE

- Amps per blower vs. baseline
- Blower runtime vs. cars washed
- Wet-car complaints by hour of day

Rule: Fix water condition and blower placement before adding horsepower.